

Yissum Research Intelligence Report — Exact & Social Sciences — August 2026

2 paper(s) · Exact & Social Sciences · Monthly · August 2026 · Generated August 01, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

Two promising research avenues emerge from recent work at Hebrew University, offering significant commercial potential across diverse sectors. One focuses on groundbreaking advancements in catalyst durability for clean energy and industrial applications, leveraging advanced nanospectroscopy to engineer more resilient materials. The second presents innovative AI-driven behavioral interventions for online food platforms, demonstrating a novel approach to foster long-term healthy eating habits and user engagement. Both projects represent opportunities for substantial market impact and strategic partnerships.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- **Elad Gross** — Materials, Clean Energy
- **Roni Lotan** — FoodTech, Software/AI, Other

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. IR nanospectroscopy mapping of facet-dependent sulfur poisoning and thermal regeneration on platinum nanocrystals.

26/50

"Breakthrough catalyst insights promise more durable fuel cells and industrial processes, cutting costs and boosting efficiency."

Main researcher: Elad Gross · elad.gross@mail.huji.ac.il.

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Published in Faraday discussions · 2026-04

ABOUT THIS RESEARCH

Researchers used advanced infrared nanospectroscopy to map how sulfur poisons individual platinum nanocrystals at a structural level. This allows them to see exactly how different crystal surfaces react to sulfur and how heat can be used to clean them.

COMMERCIAL ANGLE

The ability to map surface-level poisoning provides a roadmap for engineering more durable and efficient catalysts for fuel cells and industrial chemical processes.

COMMERCIALISATION METRICS

Novelty

7/10

The research offers significant methodological novelty by applying infrared nanospectroscopy to map facet-specific poisoning on individual nanocrystals, moving beyond bulk characterization. While the application to sulfur poisoning is a known industrial challenge, the high-resolution spatial mapping of adsorption geometry provides new, granular insights into catalyst degradation mechanisms.

Commercial Potential

5/10

While the research provides high-value fundamental insights for catalyst design, it is a characterization study rather than a novel material or a proprietary process with direct commercial application. The findings could inform future R&D for industrial catalysis, but the work itself lacks a clear path to a standalone product or licensable technology.

Market Size

7/10

The research targets a fundamental bottleneck in catalysis (sulfur poisoning) that affects massive industrial markets, including fuel cells, hydrogen production, and petrochemical refining. While the underlying science is highly relevant to high-value industrial sectors, the specific focus on nanoscale characterization is more of an enabling diagnostic tool than a direct commercial product.

Tech Readiness

3/10

The research is at a fundamental mechanistic stage, using advanced spectroscopy to resolve nanoscale structure-reactivity relationships rather than testing a prototype device or industrial process. While scientifically significant for catalyst design, it remains in the basic research/proof-of-concept phase (TRL 2-3) aimed at informing future materials engineering.

IP Strength

4/10

The work describes fundamental scientific characterization and mechanistic insights rather than a novel composition of matter, process, or device that is immediately patentable. While the findings could inform the design of more durable catalysts, the research itself is observational and lacks a proprietary technical solution or platform application.

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2. Beyond one-shot nudges: the effects of repeated interventions for healthier online meal ordering.22/50

"AI-driven nudges for food platforms ensure sustained healthy choices, unlocking new revenue and user loyalty."

Main researcher: Roni Lotan

School of Public Policy, The Hebrew University of Jerusalem, Jerusalem, Israel.; For a Change, Research Lab, Tel Aviv, Israel.

Published · 2026-05

ABOUT THIS RESEARCH

The study investigates how different digital nudges, specifically real-time feedback and assortment changes, influence healthy food choices in online ordering platforms over multiple days. It finds that combining structural changes with moment-of-choice feedback prevents the common 'decay effect' where users revert to old habits, maintaining higher health scores over time.

COMMERCIAL ANGLE

Developing specialized UI/UX behavioral optimization software for food delivery and grocery platforms to drive long-term consumer adherence to nutritional guidelines.

COMMERCIALISATION METRICS

Novelty4/10

The research applies well-established behavioral economics principles (nudging and assortment) to a known domain, offering incremental insight into temporal decay rather than introducing a novel mechanism or disruptive technology. While the longitudinal focus adds value to digital health strategy, it does not represent a fundamental scientific breakthrough or a new platform-enabling methodology.

Commercial Potential6/10

The research identifies a specific, implementable UX strategy (combining assortment and feedback) that has direct B2B licensing potential for food delivery platforms looking to improve ESG metrics or user health outcomes. However, it lacks a proprietary technical moat or deep-tech component, functioning more as an optimization playbook for existing digital interfaces.

Market Size7/10

The target market is massive, encompassing the global digital food delivery and e-commerce sectors where even marginal improvements in consumer choice behavior drive significant commercial value. However, the application is a feature-level optimization rather than a standalone platform, limiting its direct capture of the total addressable market.

Tech Readiness3/10

The technology is at a fundamental/proof-of-concept stage, utilizing a mock platform and hypothetical orders rather than a live commercial environment. While the behavioral insights are promising, the study lacks integration into real-world transactional systems or large-scale user environments.

IP Strength2/10

The innovation relies on behavioral economics principles (nudging, assortment, and feedback) which are widely documented in academic literature and are generally considered non-patentable psychological interventions. There is no evidence of a proprietary algorithm, novel software architecture, or unique hardware that would provide a defensible technical moat.

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